UPGRADE PACK · DESIGN BOOK · 7 NOV 2026

PROTOCOL *EXPLORER* · NEXT

Designs for six new lessons, nine new features and one inspector fix, in the app's own visual language. Every board pairs a mockup with the rules it has to follow; the specs in **docs/** and data in **data/** carry the details.

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NEW LESSONS

HTTP/3 over QUIC · WebSocket · IPv6 & Neighbor Discovery · Ethernet & Switching · Wi-Fi & WPA3 · DNS over HTTPS + DNSSEC

NEW FEATURES

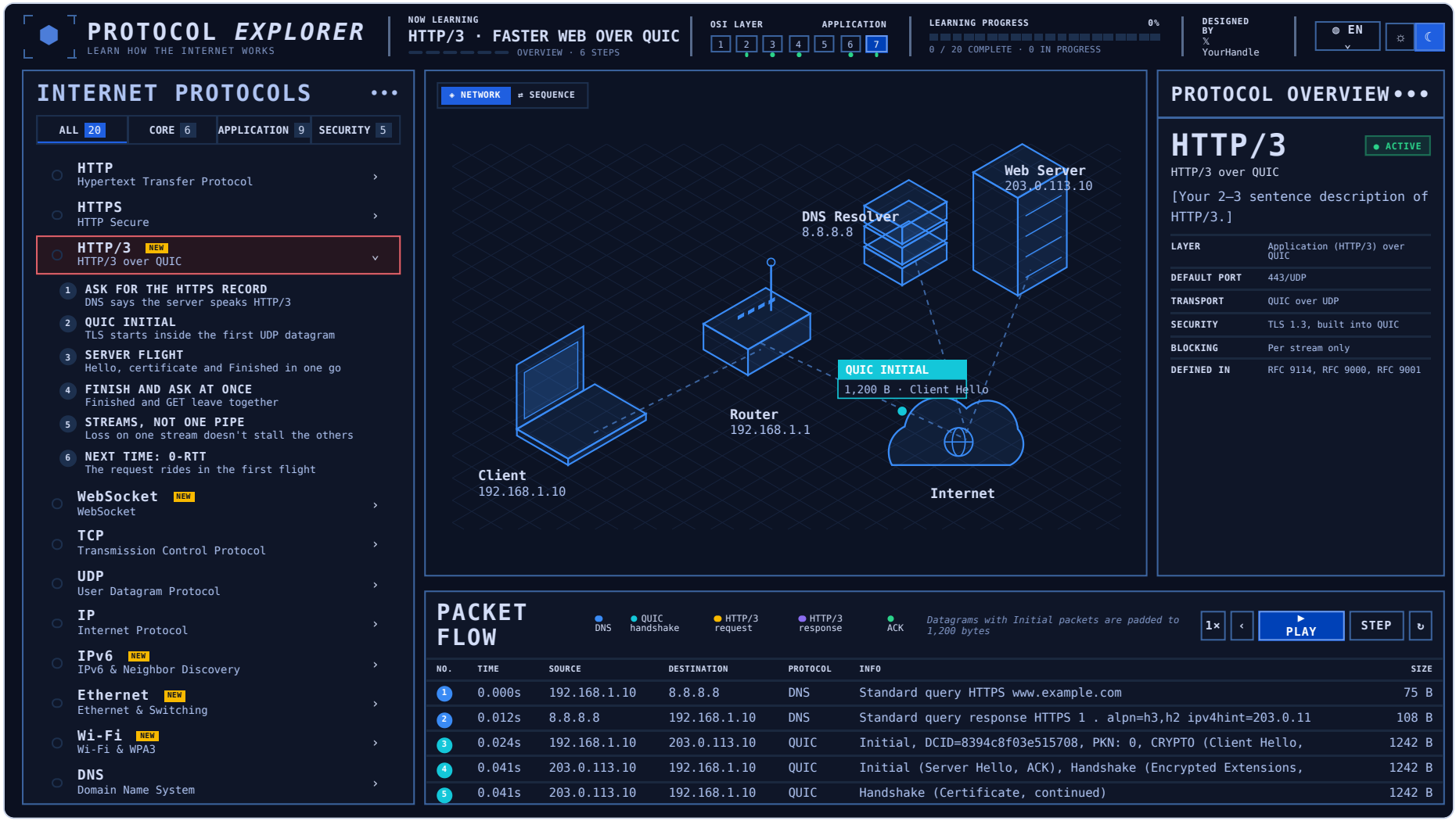
Deep links · What-if lab · Compare view · Quiz · Packet filter · Glossary · Capture import/export · Shortcuts overlay · Offline

RULE

Nothing here may change the parity view: with `?parity=1` the app must still pass the reference comparison exactly.

Lesson library · 20 lessons

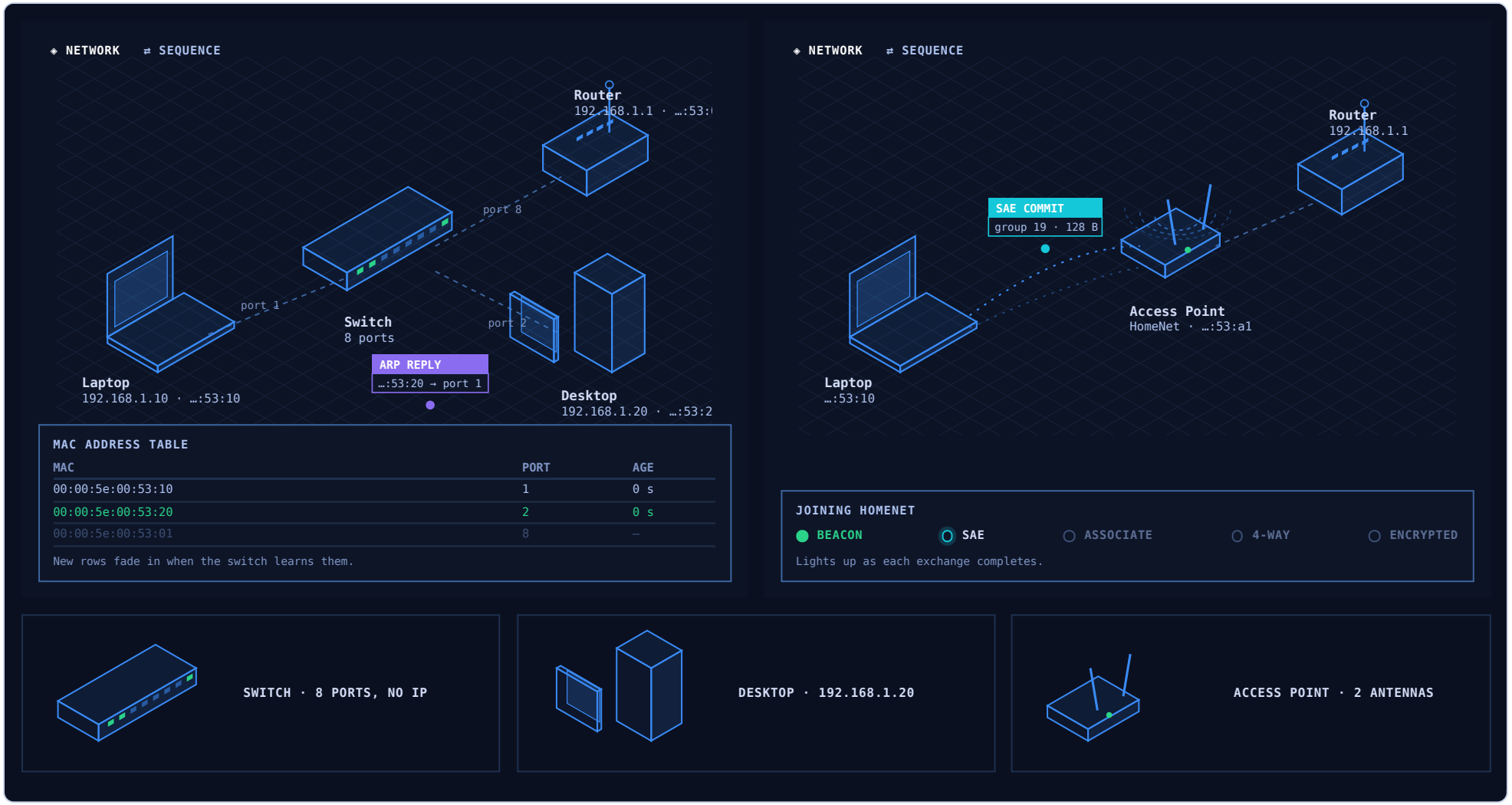
Wide layout (1672 canvas). Six new lessons join the drawer with a NEW badge; tab counts update; the OSI strip marks layers that have lessons with a green dot.



1. Drawer order: HTTP, HTTPS, **HTTP/3**, **WebSocket**, TCP, UDP, IP, **IPv6**, **Ethernet**, **Wi-Fi**, DNS, **DoH**, DHCP, ICMP, SMTP, POP3, IMAP, FTP, SSH, TLS/SSL.
2. Tabs: ALL 20 · CORE 6 (+IPv6, Ethernet) · APPLICATION 9 (+HTTP/3, WebSocket) · SECURITY 5 (+Wi-Fi, DoH).
3. **NEW** badge: 8.5px/800 on amber, shown until the lesson is opened once (store in progress).
4. OSI coverage dots under squares with at least one lesson (2, 3, 4, 6, 7). Hover a square: tooltip listing its lessons.
5. Learning progress counts out of 20.
6. In parity mode (`?parity=1`) all of this is hidden and the app is the original 14-lesson replica.

New devices and the two local-network lessons

Ethernet (left) needs a switch and a second computer; Wi-Fi (right) needs an access point and radio links. Each lesson adds one small live card in the simulation panel.



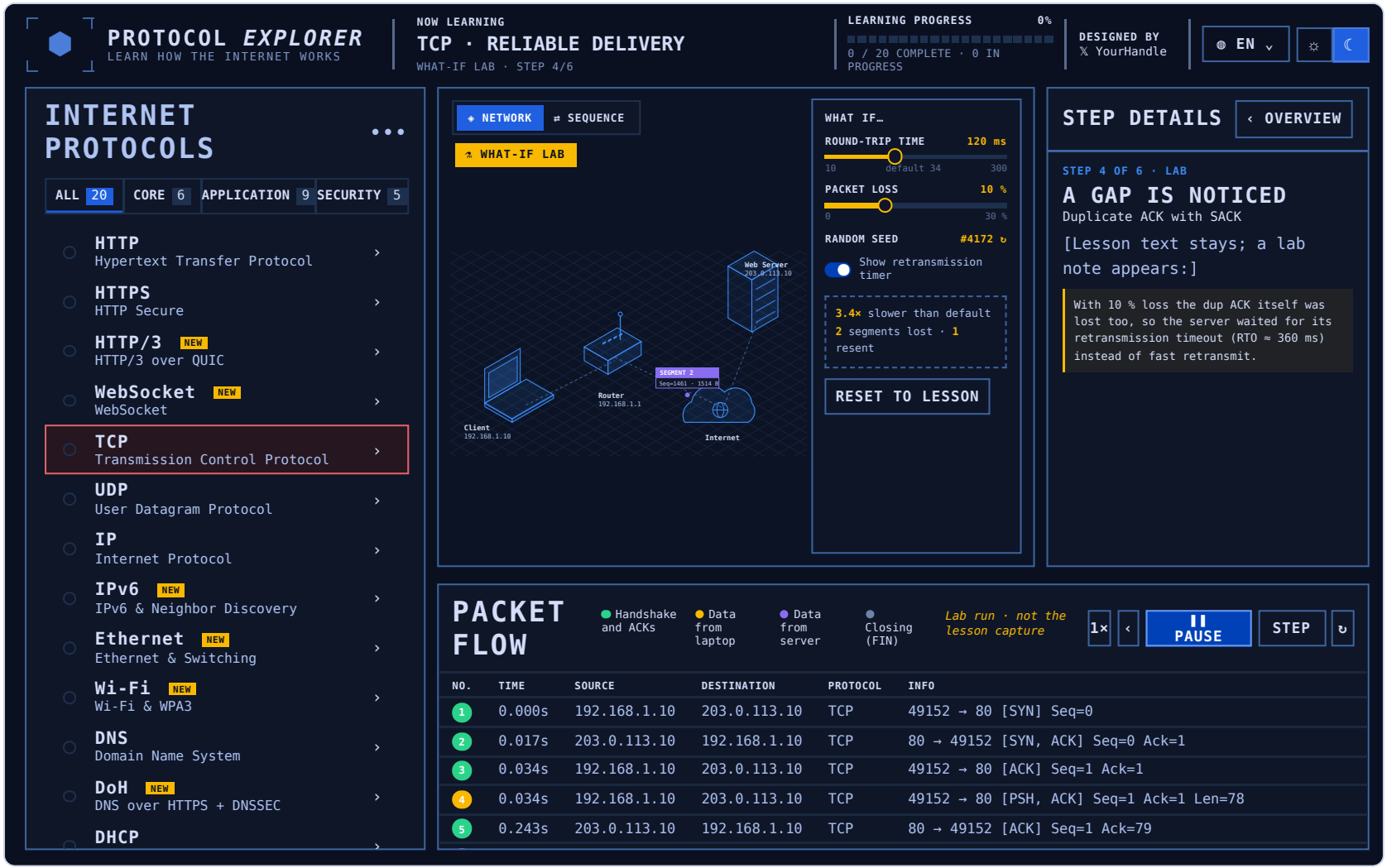
1. **Switch**: flat isometric box with 8 port LEDs; the port a frame uses lights green. No IP label: "Switch / 8 ports".
2. **Desktop**: tower + monitor, 192.168.1.20. **Access point**: low box with two antennas and a status LED.
3. Port labels sit next to each link (port 1, 2, 8) so learning is visible.
4. **MAC table card** (Ethernet): rows fade in when the switch learns an address; unknown entries are ghosted.
5. **Radio links** (Wi-Fi): dotted curves, not dashed lanes; waves pulse from the access point while beacons play.
6. **Joining card** (Wi-Fi): five stages light up as each exchange completes.

Same engine and table as every lesson. QUIC chips use the new teal colour; a small comparison chip appears once the GET is sent.



1. Packet chip "QUIC INITIAL" with a 1,200 B note: the padding is a teaching point.
2. Step Details uses the lesson spec in `data/new-lessons.json`; the explanation is written fresh.
3. "vs HTTPS: GET 52 ms sooner" chip appears at step 4: computed from both lessons' packet times, not typed.
4. Inspector: QUIC long-header fields (form, version, DCID/SCID, token, length, packet number) and CRYPTO/PADDING frames, with notes, like every other layer.

A lab panel inside the simulation re-runs the current lesson with different latency and loss. Available for TCP, UDP, HTTP, HTTPS and HTTP/3.



1. Open with L or the ••• menu. An amber "WHAT-IF LAB" badge marks that you are no longer watching the lesson capture.
2. Sliders: round-trip time 10–300 ms, packet loss 0–30 %. A seed makes runs repeatable; ↻ picks a new one.
3. The engine recomputes times and drops packets with the seeded RNG; TCP retransmits (fast retransmit or RTO), UDP just shows gaps.
4. Rows get LOST / RESENT tags; the legend shows "Lab run · not the lesson capture".
5. Result box: slowdown vs the lesson, packets lost and resent. A lab note appears in Step Details when the outcome differs from the lesson.
6. "Reset to lesson" restores the exact reference packets.

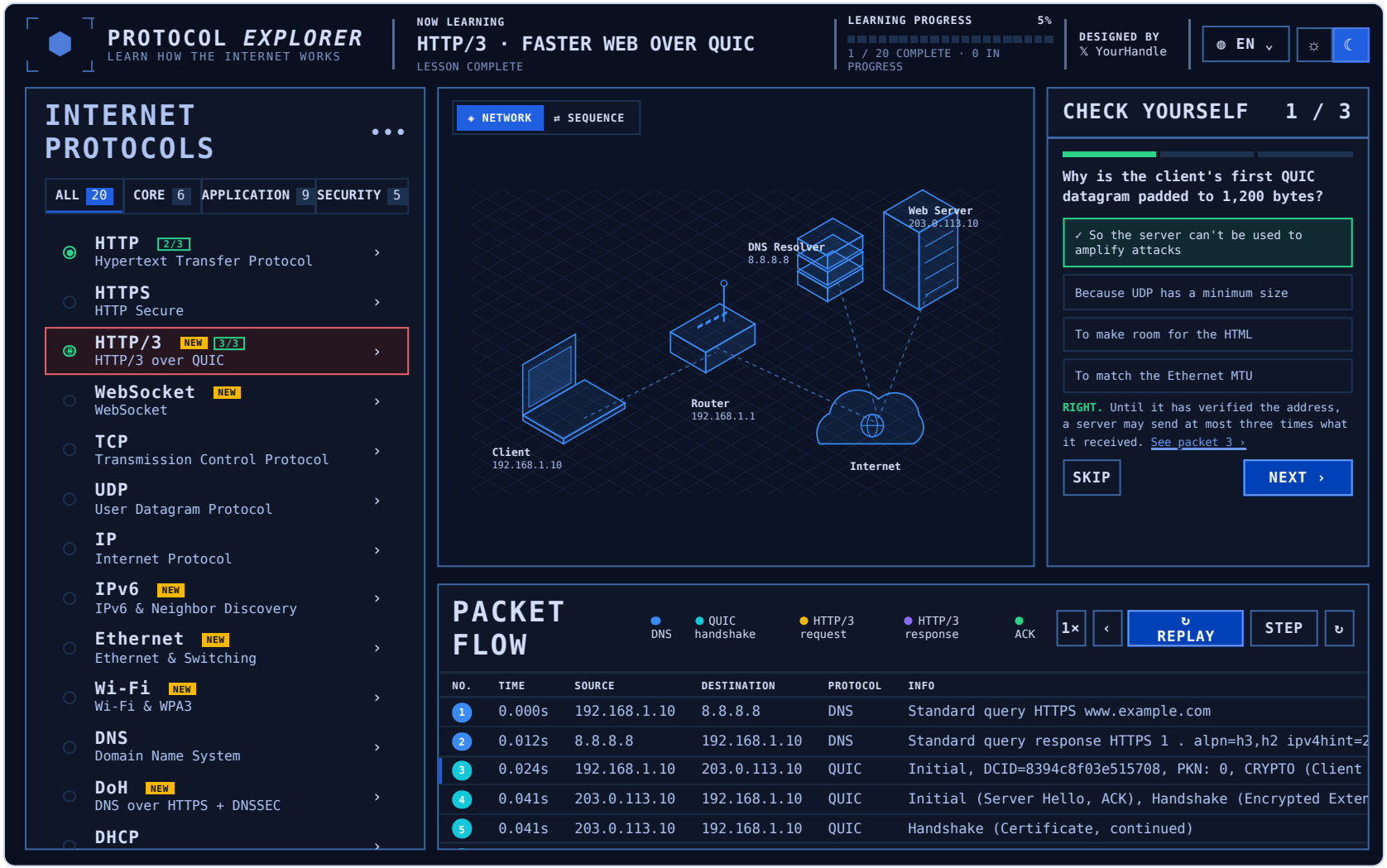
Compare lessons

A third view next to Network and Sequence: two or three lessons on one time axis, so the cost of each handshake is visible.



1. Pick lessons with chips in the side panel; HTTP, HTTPS and HTTP/3 is the default set.
2. Bars come from each lesson's packet times: DNS, TCP, TLS or QUIC handshake, then request to first byte.
3. Amber marker: when the GET leaves. Green marker: first response byte.
4. Side panel: round trips, GET time, first byte, encrypted or not, packet count, plus one generated takeaway sentence.

After the summary, three questions per lesson (`data/quiz.json`). Answers link back to the packet that explains them.



1. Starts from "Check yourself" on the Lesson Summary; one question at a time with a progress bar.
2. Options shuffled; the chosen answer turns green or red, then the explanation and "See packet N ›" (opens the inspector).
3. Score stored with progress; the drawer shows 3/3 next to the lesson and fills its ring.
4. Keyboard: 1–4 choose, Enter next.

A filter in the packet-flow header and dotted-underlined terms in explanations.



1. `/` focuses the filter. Free text matches any column; simple expressions: `proto==tcp`, `ip==8.8.8.8`, `len>100`, combined with `&&`.
2. Non-matching rows dim (numbering stays); the counter shows "3 of 11".
3. Glossary terms (`data/glossary-en.json`) are underlined on first use; hover or focus shows a card with the definition and "Seen in packet N ›".

Open your own capture

Load a .pcap/.pcapng file and explore it with the same table and inspector. Parsing happens in the browser.



1. Entry: ••• → "Open a capture…" or drag a file anywhere onto the page.
2. Privacy line is mandatory: nothing is uploaded.
3. Your capture mode: banner with file name, count and duration; table and inspector work as in lessons; no animation.
4. Decoders reuse the lesson encoders in reverse; unknown layers fall back to raw hex with "not decoded".
5. "In this capture" lists protocols found and links to the lessons that explain them.
6. Limit 50 MB / 100,000 packets; virtualise the table.

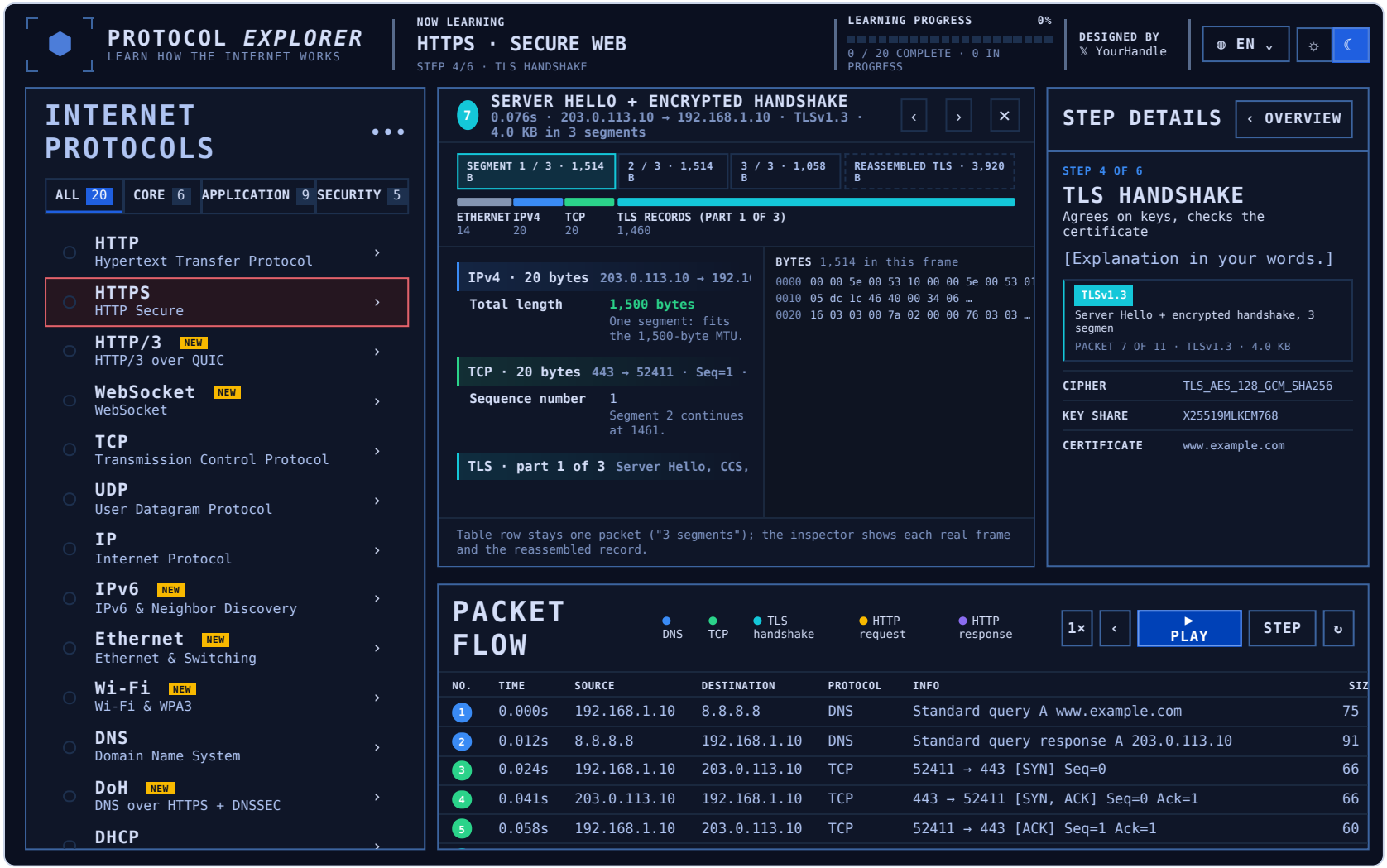
Small features that make the app easier to use and share.



1. The sidebar ••• opens a menu: copy link, compare, lab, open capture, export .pcap, keyboard shortcuts.
2. Deep links: `#http3`, `#http3/step-4`, `#http3/packet-8` (opens the inspector), `#http3/sequence`. Back and forward work.
3. `?` opens the shortcut list.
4. Toasts for "Link copied" and, on first install of the service worker, "Available offline".

Fix: real segments in the inspector

Rows like "Server Hello + encrypted handshake, 3 segments" stay one row, but the inspector shows each real frame.



1. Segment tabs: one per TCP segment, each a valid ≤1,514-byte frame with correct IPv4 total length, sequence numbers and checksums.
2. Last tab: the reassembled TLS record(s), so the TLS fields read as one message.
3. Applies to HTTPS #6–7, TLS #4, SMTP/POP3/IMAP Server Hello and DoH #7.
4. The table row, its size and the packet test stay unchanged.

Phone: lab as a bottom sheet

On phone the lab card becomes a bottom sheet above the transport bar.



1. The ⚗ button replaces reset in the bottom bar while the lab is available for the lesson.
2. The sheet keeps the two sliders, the result line and Reset to lesson.
3. Swipe down or Esc closes it; the lab badge stays on the simulation.

APPENDIX A Colours and badges added by the upgrade

TOKEN	DARK	LIGHT	USE
`--teal`	#14d1e5	#0e95a3	QUIC/TLS handshake packets (already a reference packet colour), capture banner
NEW badge	#f8b900 on #0b1020 text	same	new lessons until opened
Lab accent	--amber	#e19b00	lab badge, sliders, result numbers
Lost / resent tags	#e2626b · amber	#d6404c · amber	lab rows
Quiz correct / wrong	--green · #e2626b	--green · #d6404c	option states
Glossary underline	1px dotted --active-border (#6c99ea dark / #1f5fe0 light), 3px offset		first use of a term

Type, spacing, panels and buttons reuse the existing tokens. Every new component needs a light-theme version: derive it from the same token, never hard-code a dark colour.